

Siddhesh Bhurke

Senior Analytics Consultant | AI Solution Engineer

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SUMMARY

Senior Analytics Consultant with 7+ years driving revenue growth and operational efficiency through data-driven solutions, machine learning, and Generative AI across global supply chain networks. Deep expertise in agentic AI workflows, LLM orchestration, RAG systems, and production ML deployment. Proven track record of translating complex AI architectures into measurable business outcomes — from **11.3% refund** cost reduction with multi-agent systems to **\$12M revenue impact** through MLOps-grade production deployment. Delivered **on-site in London** for Google's EMEA analytics team; experienced working across US, Europe, Japan, and India markets. Strong stakeholder management and cross-functional collaboration with senior business leaders globally.

TECHNICAL SKILLS

Languages & packages: Python, PySpark, SQL, R, Tableau, **PowerBI**, Pandas, Keras, scikit-learn, **LLMs**
Frameworks and Tools: **Azure** Datafactory, **DataBricks**, Git, CI-CD, GCP, BigQuery, AWS, MLOps, **Agentic Frameworks**
Machine Learning: Regression, Classification, Data Analysis, Tree-Based Modeling, Generative AI, **Agentic AI**
Certifications: **Claude Certified Architect (Anthropic, 2026)** | Google Cloud Certified (2020)

PROFESSIONAL EXPERIENCE

Fractal Analytics

April 2020 –Present | [Mumbai, India](#)

Senior Analytics Consultant

Nov 2024 – Present

Refund Processing Automation (Agentic AI Decisioning) (Albertsons)

- Built a 4-agent LLM decisioning system for Jewel-Osco's refund operations — EntityProfiler, ContextAnalysis, RiskSignal, and Decision agents — delivering **11.3% refund** cost reduction, 28% productivity savings, 89.16% decision consistency, and 94.30% primary-driver reliability across a 500-case PoC (5 stores).
- Engineered 25+ KPIs across household, order, and item dimensions; converted all signals to percentile scores to enable contextual LLM reasoning. Used GPT-o3 as the decision layer for multi-agent synthesis with structured JSON outputs.
- Implemented human-in-the-loop override mechanism with audit trail; rep override rate dropped from ~40% to < 15% over 6 weeks, demonstrating system trust calibration.

SKU Segmentation Engine (POC) (Brenttag)

- Built an automated monthly SKU x Plant segmentation engine for 4 EU plants using rolling 12-month behavioral profiles (CV, trend slope, order frequency, margin) + PCA + K-means (K=5); achieved Adjusted Rand Index of 0.81 month-over-month stability and **88% planner validation** agreement.
- Automated export pipeline to AWS S3 + Power BI, replacing 3 planner-days of manual work per month.

Demand Forecasting & Reorder Point Analysis (Metro Brands)

- Built a hybrid Prophet + Gradient Boosting demand forecasting system for 320+ stores and 1,000 SKUs; improved forecast accuracy by 15%, **reduced stockouts by ~12%**, and dead stock by ~8% post-deployment.
- Created 6 demand-shape clusters as structured model features; deployed automated reorder point calculations for top 200 SKUs (MAPE <20%) with phased rollout strategy.

Analytics Consultant II

Oct 2021 – Oct 2024

SND Simulation (P&G)

- Built a 3-layer order consolidation simulation (RDD flexibility filter + fill-rate engine + cost model) identifying **\$1M savings opportunity** across top 10 customers; pilot matched model prediction within 8% variance over 4 weeks.

Last Mile Delivery (Schneider)

- Designed a multi-interval (daily/weekly/bi-weekly) shipment consolidation recommendation system for NAM market; by optimising on the logistics cost by consolidating the shipments based on reason codes and address using fuzzy matching algorithm.

Strategy and Analytics: onsite: London, United Kingdom (Google, UK)

- Identified growth opportunities by diligently monitoring SEO key metrics, leading to data-informed decisions and boosted performance across EMEA markets with **21% Q/Q active users' growth**.
- Regularly updated and evaluated the impact of quarterly OKRs using SEO metrics such as user growth and impressions, improving predictive accuracy for future quarters. Streamlined performance tracking by identifying market-specific KPIs, setting up end-to-end pipelines, and building attribution models, leading to an enhanced understanding of market performances.

Gaming Platform Revenue Analytics (Sony PlayStation)

- Applied propensity score matching (pymatch) to construct comparable test and control groups across tournament participants and non-participants — isolating causal subscription uplift from selection bias.
- Measured pre/post tournament subscription changes within matched cohorts, quantifying an **8% revenue uplift** attributable to tournament participation.

Analytics Consultant I

Apr 2020 – Sep 2021

Cost Performance Management tool (MARS)

- Built PyTorch neural network + Integrated Gradients attribution tool providing exact, completeness-satisfying cost driver decomposition; added Streamlit scenario simulation layer enabling **5-6% potential cost savings** across multiple markets and product portfolios.

Customer Service Levels (MARS)

- Deployed CatBoost + XGBoost ensemble classifier (78%/81.4% accuracy) for 12-week order fulfillment risk prediction; contributed to **\$12M revenue impact**. Built PSI-based drift monitoring that successfully triggered retraining when supply chain disruption caused PSI to spike from 0.08 to 0.34 on a key feature.

Think & Learn Pvt. Ltd.

Aug 2017 – Jun 2018 | Mumbai, India

Business development associate, Mumbai, India

- Generating revenue through counselling prospects and converting them into sales, converting free app users to paid users using Emails, Calls and Direct sales.

EDUCATION

MBA - Data Science and Data Analytics | Symbiosis University, Pune, India | Jun 2018 – March 2020

P.G. Diploma in Management | Mumbai University, Mumbai, India | July 2015 – Jun 2017

B.Sc. - Physics | Ramnarain Ruia College, Mumbai, India | Jun 2011 – April 2014

OPEN SOURCE PROJECTS

Docu-AI-Search

github.com/BhurkeSiddhesh/Docu-AI-Search

- Built a production-grade AI-powered semantic document search engine with a FastAPI backend, React 19 frontend, and hybrid Dense+Sparse (FAISS + BM25) retrieval pipeline supporting PDF, DOCX, XLSX, PPTX, and TXT files.
- Implemented advanced RAG optimizations — LLM-based query rewriting, Cross-Encoder reranking over top-20 candidates, and an agentic researcher mode that synthesizes answers across multiple documents with source citations.
- Engineered multi-provider LLM support (OpenAI, Anthropic, Gemini, Grok, HuggingFace) and a fully local GGUF inference path (LlamaCpp) with an in-app Model Manager for 25+ open-source models — no API keys required.
- Shipped with 100+ unit/integration tests (pytest + Vitest), SQLite-backed AI response caching, $O(\log N)$ B-Tree metadata indexing, Docker deployment, and a golden dataset "needle-in-haystack" benchmark for retrieval accuracy validation.

File-Forge

github.com/BhurkeSiddhesh/File-Forge | **Live: file-forge.onrender.com**

- Built and deployed a web-based multi-agent file manipulation platform (FastAPI + Vanilla JS) with AI-driven OCR and layout analysis using PaddleOCR/PPStructure — supporting PDF password removal, high-fidelity PDF-to-DOCX conversion, page extraction, and image operations (HEIC→JPEG, resize, crop).
- Designed a chained Workflow Builder that sequences multiple file operations in a single pipeline with real-time progress streaming via Server-Sent Events (SSE), reducing multi-step file processing from manual effort to a single submission.
- Deployed to production on Render with Docker, integrated CodeRabbit automated PR reviews, and built a full test suite (conftest fixtures, integration + unit tests) covering all API endpoints and edge cases.